

GAIRA V6 — Hierarchical Semantic Rebuild

Rebuilding the interpretation hierarchy above a frozen Raman foundation

scope	In-domain pure Raman only. No Ag-SERS work in this pass.
frozen	Atlas 09ed804a40836f4a05a91ba10900cded · assets/foundation byte-identical · NMF, preprocessing, NNLS projection and component registry all unchanged.
built	A leakage-free MSS layer (17 motifs), a chemical-theme layer (13 themes) derived FROM MSS, and a Pareto study justifying the number of themes.
deferred	Biological-state themes — they need functional evidence a static spectrum does not carry.

What V6 is

V6 does not tune the model. It removes a **circularity**. In V1, 25 % of every component→motif weight was copied from the component→theme matrix, so a themes-from-MSS hierarchy would have been predicting a quantity partly built out of themes. V6 rebuilds MSS from spectroscopy alone, then derives themes as groupings of motifs — a composition of two non-negative linear maps, **theme = $T^T M^T$ coord**.

Headline results

quantity	V1	V6
MSS theme leakage removed	13.4% of the mean raw score; 15 of 70 edges depended on it	eliminated
MSS band fidelity	0.594 → 0.633	improved for 12/13
MSS confidence	constant 0.33 (a NumPy bool-OR artefact)	0.475 – 0.723
Motif exemplar coverage	35.9 % of the corpus	98.8 %
Motifs	13	17
Chemical themes	11 biochemical + 2 non-biochemical (asserted)	13 (Pareto-selected from 70 partitions)
Theme top-1 / top-3	0.629 / 0.805	0.614 / 0.890
Recoverability κ	not measured	0.538 (null 0.163)

1 • Methods

1.1 The frozen foundation (unchanged)

Every spectrum is cropped to 450–1800 cm^{-1} , baseline-corrected (AsLS), smoothed (Savitzky-Golay), resampled onto a 676-bin 2 cm^{-1} grid and L2-normalised, then fitted onto the frozen 24×676 non-negative basis **H** by non-negative least squares with the dictionary held fixed. The activations are L1-normalised so a coordinate reads as a share of the reconstructed evidence. V6 touches none of this.

$$\text{coord}(x) = \text{NNLS}(\text{preprocess}(x), H) , \sum_j \text{coord}_j = 1 , \text{coord} \geq 0$$

1.2 MSS — a spectroscopy-only layer

For motif m and component j , four evidence lines, none of which can reference a theme:

```
band_jm = |{b ∈ bands(m) : ∃p ∈ peaks(j), |p-b| ≤ 16}| / |bands(m)|
cosine_jm = ( H_j / ||H_j|| , φ_m ) with φ_m(v) = ∑_b exp(-½((v-b)/9)²)
exemp_jm = min(1, ∑ contribution_pct of j's analytes matching exemplars(m) / 12)
pert_jm = min(1, (∑|ρ_dose| + ½·n_spike) / 3) matched to exemplars(m)
raw_jm = 0.30·band + 0.30·cosine + 0.30·exemp + 0.10·pert
keep if raw ≥ 0.12 · cap 6 contributors · normalise so ∑_j M_jm = 1
```

The **basis-spectrum cosine** is new in V6 and is the most direct spectroscopic evidence available: it asks whether the component's actual frozen basis spectrum looks like the motif's declared band pattern, rather than only whether discrete peak positions coincide.

The perturbation term is weighted low (0.10) because the perturbation corpus is purine-heavy. An ablation without it is always reported: the component-weight cosine between V6 and V6-without-perturbation is **0.983**, so the functional term is not driving the motif definitions.

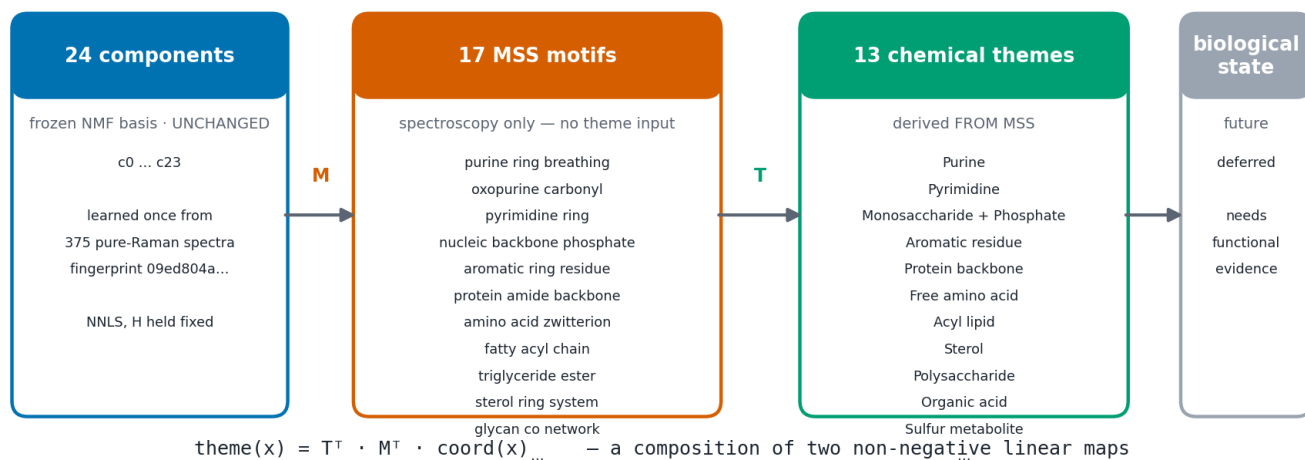
1.3 Chemical themes — derived FROM MSS

A chemical theme at level K is a **partition** of the 17 biochemical motifs into K groups. Theme composition is the sum of the member motifs' activations, so the full chain is a composition of two non-negative linear maps:

$$\text{mss}(x) = M^T \text{coord}(x) , \text{theme}(x) = T^T \text{mss}(x) = T^T M^T \text{coord}(x)$$

Chemical themes are **biochemical classes**, not disease biology. Biological-state themes are deliberately out of scope.

The V6 semantic hierarchy



V1 fed the component→theme matrix into every MSS weight (25 % of the raw score), so deriving themes from MSS would have been circular. V6 removes that term.

Figure 1 — The V6 semantic hierarchy.

2 · The audit that made V6 necessary

The V1 MSS layer computed each component→motif weight as $\text{raw} = 0.40 \cdot \text{band} + 0.35 \cdot \text{exemplar} + 0.25 \cdot \text{theme}$, with $\text{theme} = \text{ontology.W}[j, \text{parent_theme}]$ (src/gaira/engine/mss.py:195-196).

code site	statement
src/gaira/engine/mss.py:190	ti = self.onto.theme_index(parent)
src/gaira/engine/mss.py:195	theme = float(self.onto.W[j, ti])
src/gaira/engine/mss.py:196	raw = wb*band + we*exemplar + wt*theme <-- 25% from themes
src/gaira/engine/mss.py:209	evidence_breadth includes the theme indicator
src/gaira/engine/mss.py:169	purine_motif = parent_theme == 'nucleic_purine' (perturbation gate)

Part 0 — theme leakage in the V1 MSS layer

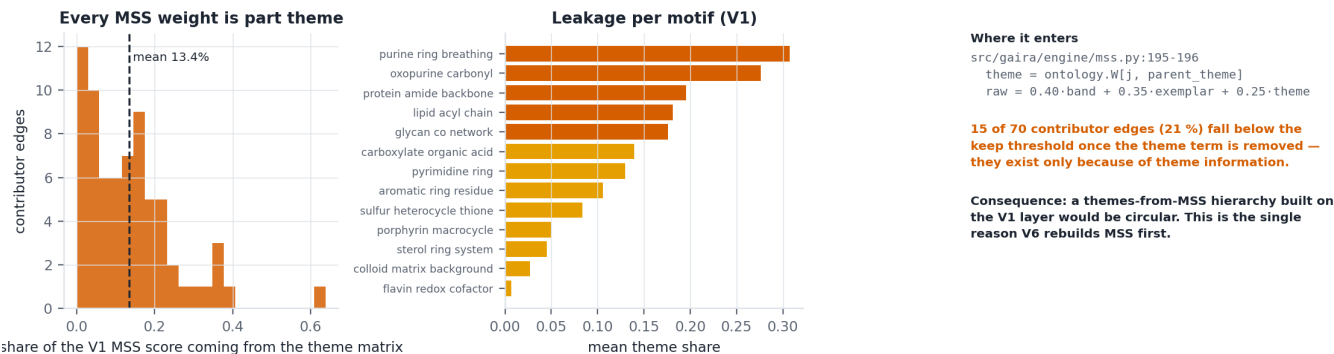


Figure 2 — Theme leakage in V1, quantified over all 70 contributor edges.

Finding

Mean theme share of the raw score **13.4%**, maximum **63.9%**. **15 of 70 edges (21 %)** fall below the keep threshold once the theme term is removed. A second defect sits alongside it: because the three evidence indicators are `np.bool_` and NumPy's `+` on booleans is logical OR, `evidence_breadth` evaluated to exactly 1/3 for every motif — V1 MSS confidence was a rescaled stability carrying no motif-discriminating information.

3 · Part 1 — MSS rebuilt without theme evidence

Part 1 — MSS rebuilt without theme evidence: quality improves, support is preserved

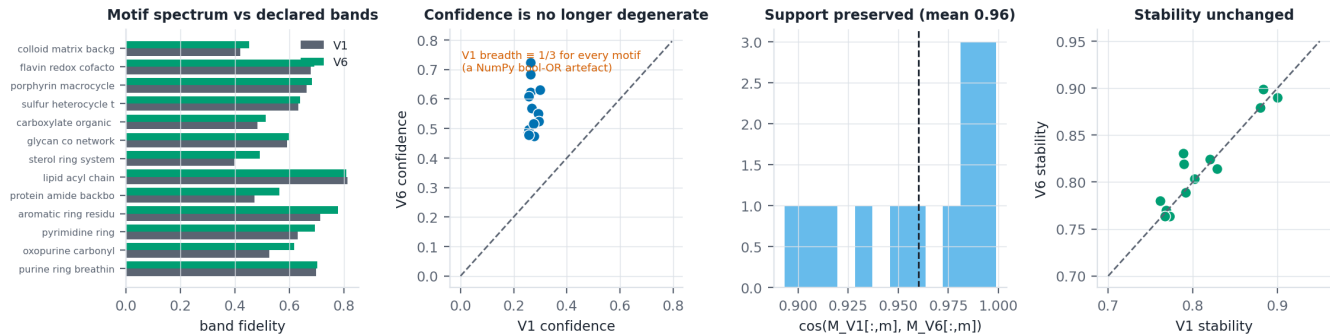


Figure 3 — Removing the theme term improves band fidelity and restores a discriminating confidence, while leaving component support and stability essentially unchanged.

motif	band fidelity V1	band fidelity V6	Δ
purine ring breathing	0.698	0.701	+0.004
oxopurine carbonyl	0.527	0.617	+0.091
pyrimidine ring	0.630	0.693	+0.063
aromatic ring residue	0.714	0.779	+0.066
protein amide backbone	0.472	0.563	+0.092
lipid acyl chain	0.814	0.809	-0.005
sterol ring system	0.398	0.491	+0.093
glycan co network	0.592	0.597	+0.005
carboxylate organic acid	0.482	0.513	+0.030
sulfur heterocycle thione	0.632	0.639	+0.007
porphyrin macrocycle	0.663	0.681	+0.018
flavin redox cofactor	0.679	0.692	+0.013
colloid matrix background	0.420	0.451	+0.031

Verdict on Part 1

Band fidelity improved for **12 of 13** motifs (0.594 → 0.633); stability is unchanged (0.812 → 0.817); component-weight cosine V1↔V6 is 0.960 and the activation Spearman is 0.887. **MSS quality improves and its support is preserved** — V6 is a purification, not a replacement.

4 · Part 2 — auditing and redefining the motifs

Part 2 — auditing the motifs, and what V6 changes

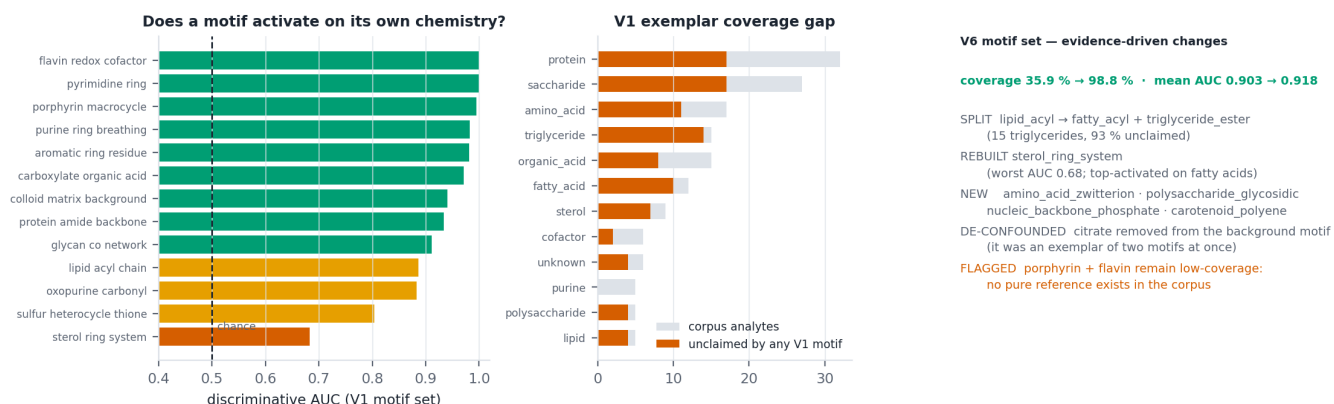


Figure 4 — What the V1 motif set could and could not describe.

Three findings drove the V6 motif set:

- **Coverage.** V1 exemplar lists named only **35.9 %** of the 167 corpus analytes. Whole families were unclaimed: triglyceride 93 %, fatty acid 83 %, polysaccharide 80 %, sterol 78 %, nucleic acid 100 %, phospholipid 100 %, carotenoid 100 %.
- **Discrimination.** sterol_ring_system had the worst AUC of any motif (0.683) and its top-activating family was fatty_acid, not sterol — it did not describe the chemistry it named.
- **Redundancy.** The two most overlapping motif pairs were porphyrin↔flavin (component-support cosine 0.699 — both borrow, because the corpus contains no pure reference for either) and carboxylate↔background (0.687 — citrate was an exemplar of both at once).

The V6 motif set

change	motif	evidence
SPLIT	lipid_acyl_chain → fatty_acyl_chain + triglyceride_ester	15 triglycerides and 12 free fatty acids; the ester C=O (~1745) and C-O-C (~1160) separate them
REBUILT	sterol_ring_system	re-banded on canonical cholesterol modes (548/608/702/958/1128/1670)
NEW	amino_acid_zwitterion	17 free amino acids, 65 % unclaimed; COO ⁻ 1410, C-N 1030, NH ₃ ⁺ 1520
NEW	polysaccharide_glycosidic	glycosidic C-O-C 890 β / 940 α — separates polymers from free sugars
NEW	nucleic_backbone_phosphate	PO ₂ ⁻ 1094, O-P-O 810/835 — 3 nucleic acids, several phosphometabolites
NEW	carotenoid_polyene	v ₁ C=C 1520, v ₂ C-C 1157 — resonance scatterers, previously undescribed
DE-CONFOUNDED	colloid_matrix_background	citrate removed from the exemplar list
FLAGGED	porphyrin_macrocycle, flavin_redox_cofactor	retained but marked low_coverage — no pure reference exists in the corpus

Exemplar coverage **35.9 % → 98.8 %**; mean discriminative AUC **0.903 → 0.918**; no motif below AUC 0.60.

The 17 V6 MSS motifs — implied Raman spectrum (red, $M[:,m]^T H$) against the declared band profile (blue dashed)

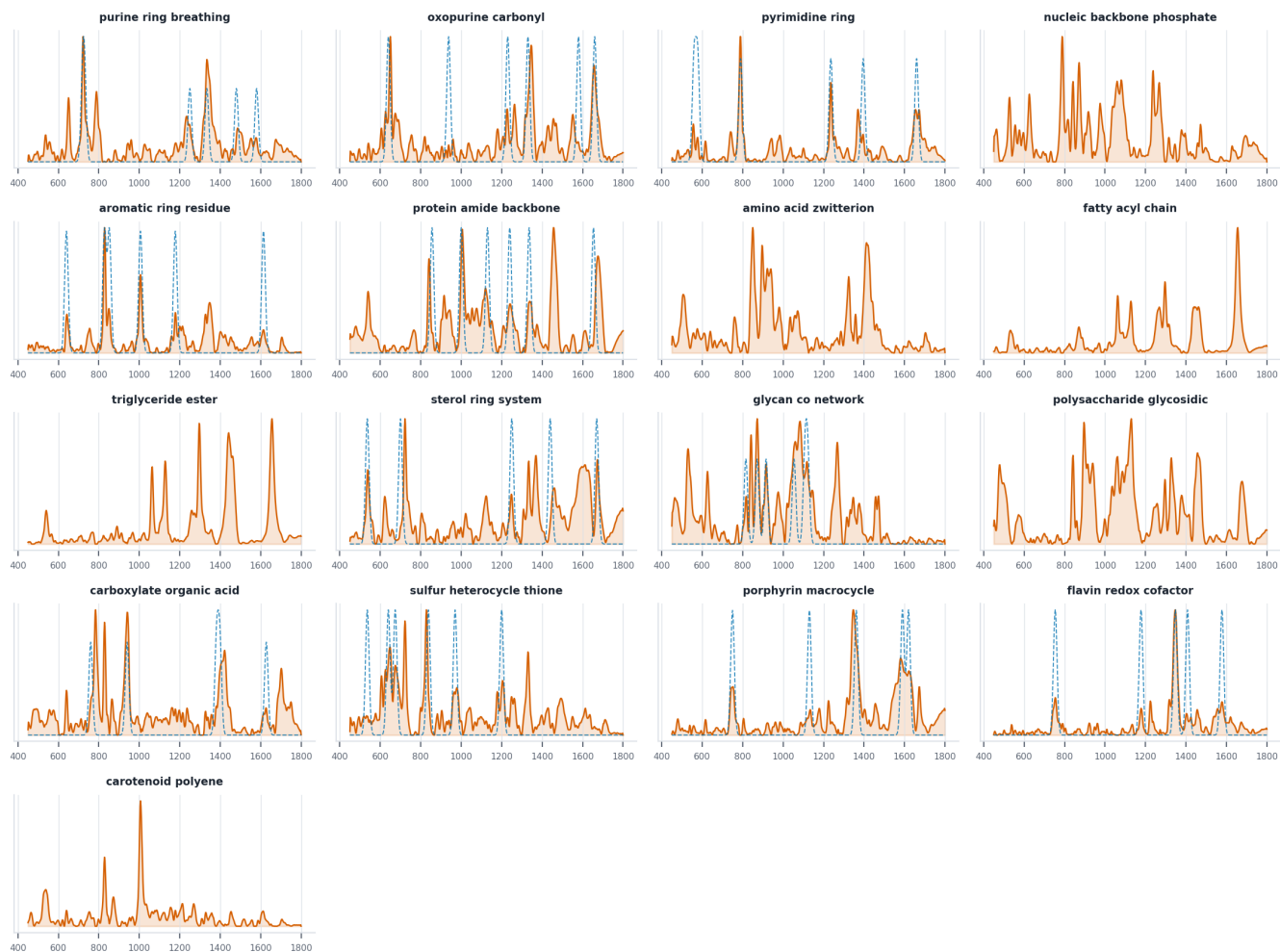


Figure 5 — The 17 V6 motifs: implied Raman spectrum (red) against the declared band profile (blue dashed).

5 · Parts 3-5 — the theme layer and its optimisation

Themes are groupings of motifs, so the number of themes is a real design choice. Sixteen levels ($K = 2 \dots 17$) $\times$ five generation methods = 70 candidate hierarchies, each scored against a 2000-fold label-permutation null.

Why raw accuracy is the wrong objective

Top-1 accuracy rises **mechanically** as K falls — a two-theme hierarchy is right half the time by guessing. The sweep therefore reports chance-corrected recoverability, and an interpretability composite that explicitly penalises trivial coarseness:

$$\kappa = (\text{top1} - \text{null}) / (1 - \text{null})$$

$$I = 0.4 \cdot C_{\text{chem}} + 0.3 \cdot C_{\text{spec}} + 0.3 \cdot \log K / \log K_{\text{max}}$$

$0.4 \cdot \text{chemical_coherence} + 0.3 \cdot \text{spectral_coherence} + 0.3 \cdot \text{resolution}$, where $\text{chemical_coherence} = 1 - \text{normalised entropy of member analytes' chemical families}$, $\text{spectral_coherence} = \text{mean pairwise cosine of member motifs' implied Raman spectra}$, $\text{resolution} = \log(K)/\log(K_{\text{max}})$. Resolution is included so that a 2-theme hierarchy is not rewarded for being trivially coherent.

Part 4 — theme optimisation: accuracy alone is the wrong objective

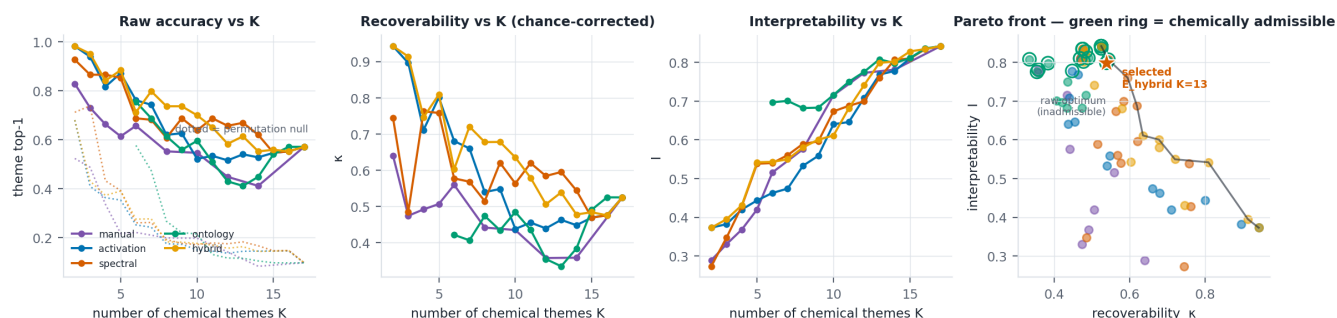


Figure 6 — Accuracy, recoverability and interpretability against K , and the Pareto front.

Breaking the tie

The composite $\kappa \times I$ is **flat** across the front (0.43-0.45 over four candidates spanning $K = 13, 14$ and 17 and three different methods) — statistically indistinguishable at $n = 163$. As with the atlas's own ICA-versus-NMF tie, the resolution is a **pre-stated constraint**, not a third decimal place:

The tie-break

A theme must span at most one non-bridging chemical superclass. Superclasses fixed on chemistry before scoring; phosphate_ester is the only BRIDGING class. Only **21 of 70** partitions qualify. The raw-score optimum (**C_spectral K=13**, score 0.4523) is **inadmissible**: it merges polysaccharide with protein backbone, a spectral-overlap artefact no chemist would propose. Within the admissible band the smallest K is preferred, because a theme layer that does not abstract is only the motif layer relabelled.

The five generation methods

method	best K	κ	I	$\kappa \times I$	top-1
A_manual	17	0.524	0.842	0.4415	0.571
B_activation	17	0.525	0.842	0.4420	0.571
C_spectral	13	0.595	0.760	0.4523	0.669
D_ontology	17	0.525	0.842	0.4418	0.571

method	best K	κ	I	$\kappa \times I$	top-1
E_hybrid	17	0.524	0.842	0.4415	0.571

A manual — an expert chemical hierarchy fixed before scoring, evaluated only at its own defined levels. **B activation** — agglomerative on motif co-activation across the corpus. **C spectral** — agglomerative on the cosine between motifs' implied Raman spectra. **D ontology** — chemical class plus shared exemplar analytes. **E hybrid** — the mean of the B/C/D distances; **selected**, because it is the only distance that sees co-activation, spectral shape and named chemistry at once.

6 · The selected hierarchy

chemical theme	member MSS motifs
Purine	oxopurine_carbonyl, purine_ring_breathing
Pyrimidine	pyrimidine_ring
Monosaccharide + Phosphate	glycan_co_network, nucleic_backbone_phosphate
Aromatic residue	aromatic_ring_residue
Protein backbone	protein_amide_backbone
Free amino acid	amino_acid_zwitterion
Acyl lipid	fatty_acyl_chain, triglyceride_ester
Sterol	sterol_ring_system
Polysaccharide	polysaccharide_glycosidic
Organic acid	carboxylate_organic_acid
Sulfur metabolite	sulfur_heterocycle_thione
Flavin + Porphyrin	flavin_redox_cofactor, porphyrin_macrocycle
Carotenoid	carotenoid_polyene

E_hybrid at K = 13 — top-1 0.614 against a permutation null of 0.163 ($\kappa = 0.538$), macro-F1 0.501, balanced accuracy 0.554, interpretability 0.799. Every one of the four merges is chemically defensible: purine ring + oxopurine carbonyl; monosaccharide + phosphate ester (sugar phosphates); fatty acyl + acylglycerol; flavin + porphyrin (both conjugated N-heterocyclic cofactors, both corpus-limited).

The two interpretive maps of V6 — neither uses a theme label as input

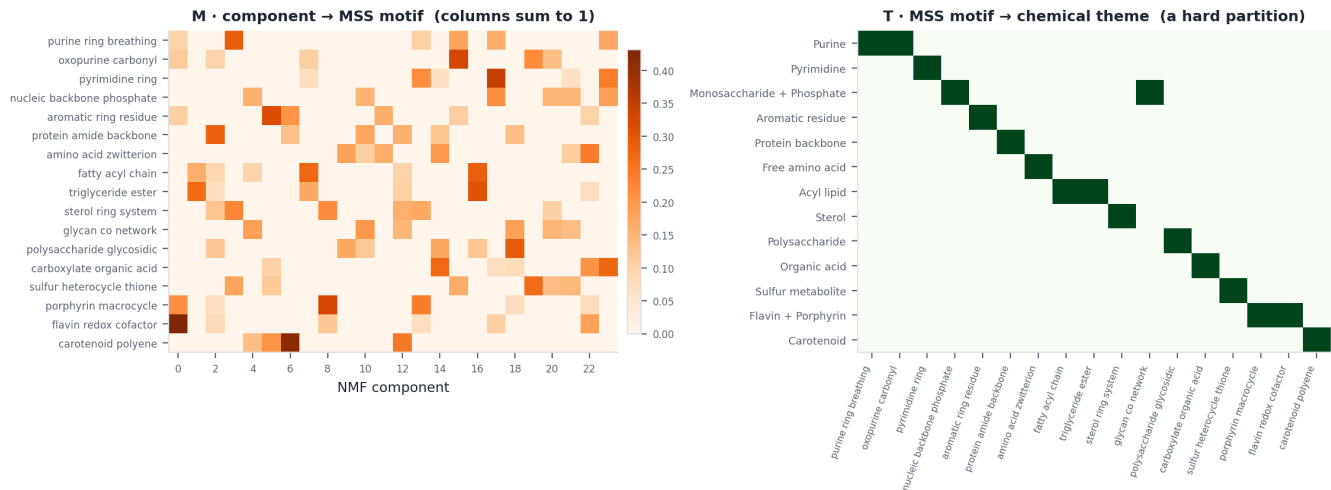


Figure 7 — M and T. Neither takes a theme label as input.

7 · Part 7 — evaluation over the full corpus

Every one of the 167 Raman grounding analytes was pushed through the frozen V6 stack; 163 carry an expected motif and are scored.

Part 7 — full evaluation over every Raman grounding analyte

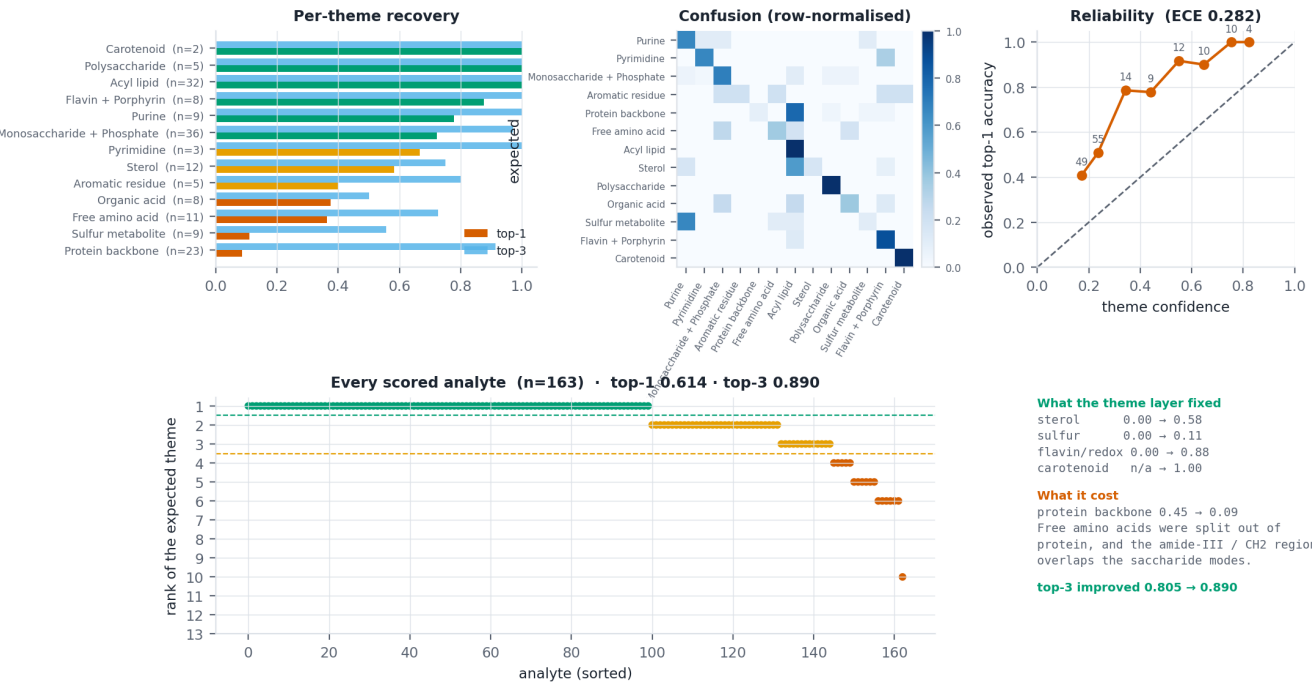


Figure 8 — Per-theme recovery, confusion, reliability and the rank of the expected theme for every scored analyte.

Per-theme performance

chemical theme	motifs	n	top-1	top-3	med rank	most confused with
Purine	2	9	0.78	1.00	1	Pyrimidine
Pyrimidine	1	3	0.67	1.00	1	Flavin + Porphyrin
Monosaccharide + Phosphate	2	36	0.72	0.97	1	Acyl lipid
Aromatic residue	1	5	0.40	0.80	2	Monosaccharide + Phosp
Protein backbone	1	23	0.09	0.91	2	Acyl lipid
Free amino acid	1	11	0.36	0.73	2	Monosaccharide + Phosp
Acyl lipid	2	32	1.00	1.00	1	-
Sterol	1	12	0.58	0.75	1	Acyl lipid
Polysaccharide	1	5	1.00	1.00	1	-
Organic acid	1	8	0.38	0.50	3	Monosaccharide + Phosp
Sulfur metabolite	1	9	0.11	0.56	3	Purine
Flavin + Porphyrin	2	8	0.88	1.00	1	Acyl lipid
Carotenoid	1	2	1.00	1.00	1	-

V6 recovers four chemistries V1 could never reach. sterol 0.00 → 0.58 · sulfur 0.00 → 0.11 · flavin/redox 0.00 → 0.88 · carotenoid n/a → 1.00. Acyl lipid, polysaccharide and carotenoid are perfect; purine 0.78; monosaccharide+phosphate 0.72. Top-3 rose 0.805 → 0.890.

The honest cost. Protein backbone fell 0.45 → 0.09. Splitting free amino acids into their own motif is chemically right, but it leaves the amide backbone competing with saccharide modes in the same 1240–1340 cm⁻¹ region, and it loses. Sulfur is recovered but only at 0.11. Calibration is loose (ECE 0.28): confidence over-states accuracy at the top of the range.

8 · Parts 8-10 — representative analytes and explainability

tier	analyte	family	expected	predicted	theme rank	motif rank
excellent	triarachidin	triglyceride	Acyl lipid	Acyl lipid	1	1
good	lauric acid	fatty_acid	Acyl lipid	Acyl lipid	1	2
moderate	(+)-galactose	saccharide	Monosaccharide + Phosphate	Purine	2	6
poor	acetyl-coa	cofactor	Sulfur metabolite	Purine	6	3
failure	ascorbate	organic_acid	Organic acid	Acyl lipid	10	13

Part 10 — one real analyte, end to end, at three performance tiers
green = expected · red = what actually won

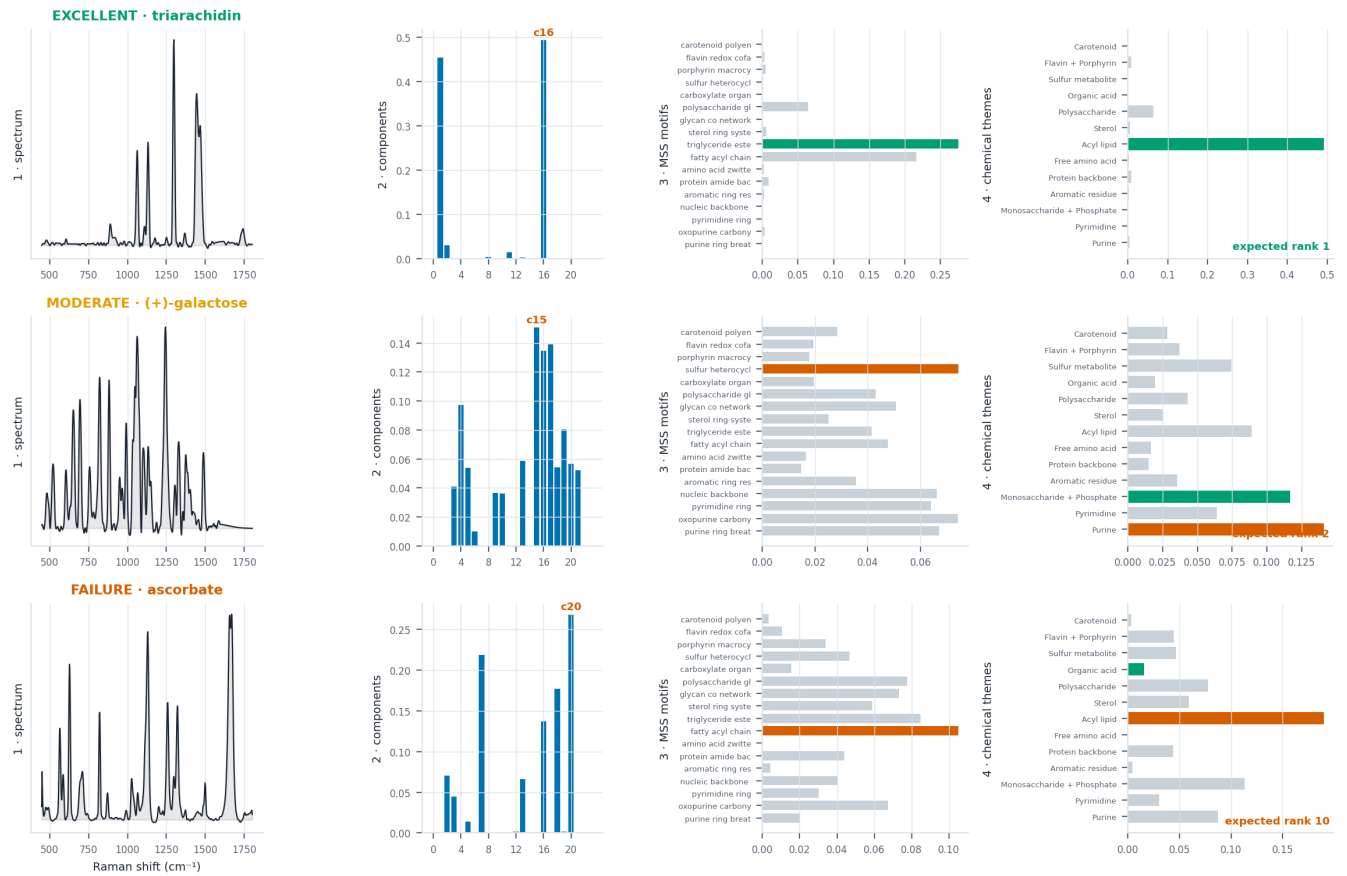


Figure 9 — One real analyte, end to end, at three performance tiers.

The three-level radar

The V1 radar showed themes only. V6 shows the whole ladder, because when a call goes wrong the level at which it went wrong *is* the diagnosis: a clean component radar with a wrong theme radar is an interpretation failure; a diffuse component radar is a representation failure. They need different fixes.

Part 9 — the radar is now three nested radars, one per abstraction level
component (what the atlas saw) → motif (what spectroscopy says) → theme (what chemistry it implies)

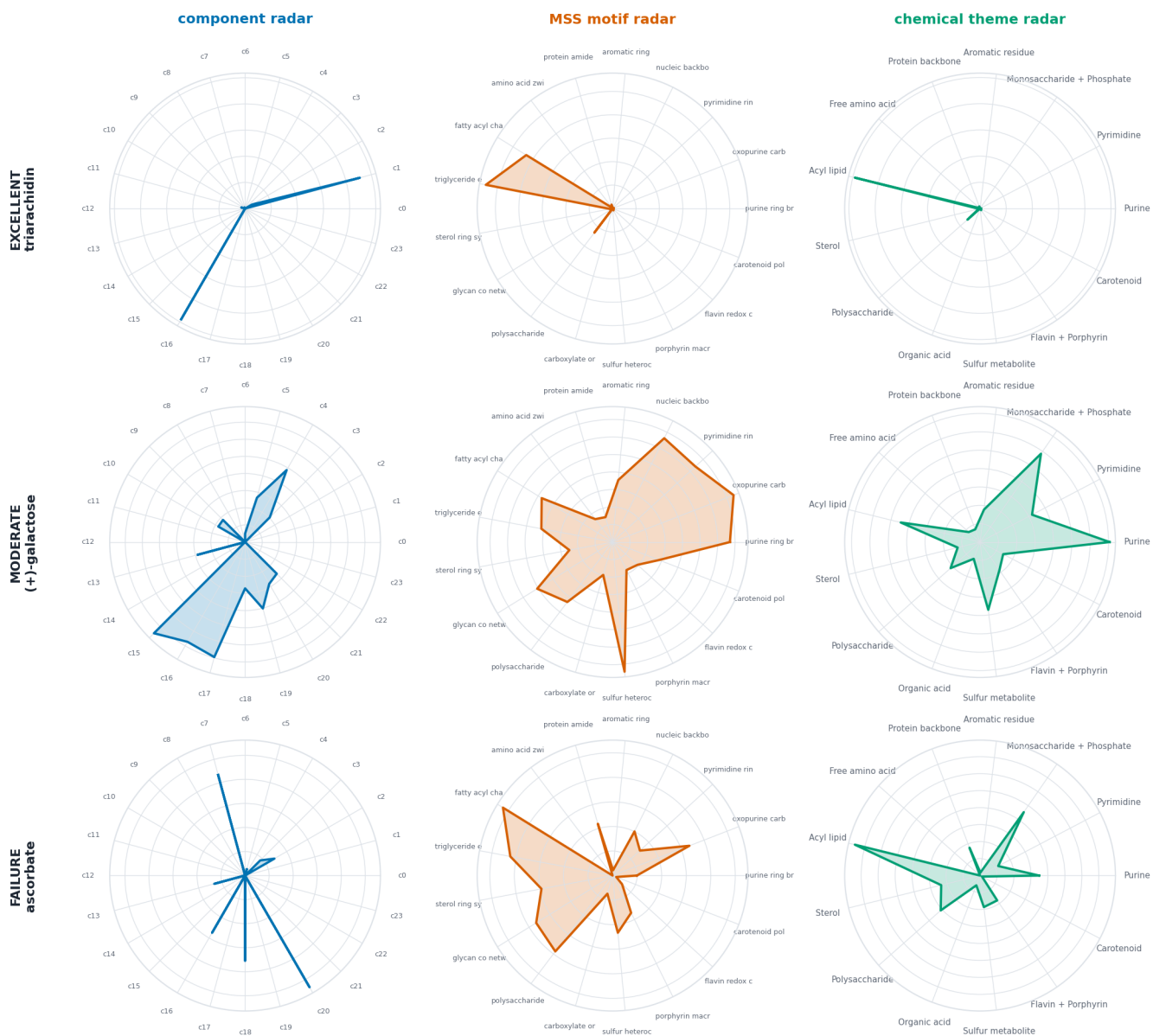


Figure 10 — Component radar → MSS motif radar → chemical theme radar.

9 · Limitations

- **Protein backbone regressed** from 0.45 to 0.09 top-1. This is the clearest cost of the V6 motif split and is not yet resolved.
- **Sterol remains weak** (0.58) even after re-banding. The frozen atlas contains no component that isolates the steroid ring system — c3 carries the v0.1 'sterol' label but its top reference loading is adenine. This is a *foundation* limit, not a motif-definition limit, and V6 cannot fix it without unfreezing the atlas.
- **Porphyrin and flavin remain low-coverage.** The corpus holds no pure porphyrin or flavin reference; both motifs borrow protein components. Their apparent 0.88 recovery rests on 8 analytes.
- **Carotenoid rests on two analytes.** Its perfect score is not meaningful evidence.
- **Calibration is loose** (ECE 0.28). Theme confidence should not be read as a probability.
- **The expected-theme map is a curated evaluation overlay**, derived from motif exemplar membership. It is not a learned layer and is used nowhere in inference, but it does mean the evaluation measures agreement with a human-defined chemistry, not ground truth in any stronger sense.
- **In-domain only.** Everything here is pure Raman on the corpus the atlas was fitted to. This bounds what the layers can do at best; it is not a generalisation test.

10 · Recommendations

- **Repair the protein backbone motif.** Its bands (856/1000/1130/1240/1336/1654) overlap the saccharide C-O region. Consider narrowing it to amide I + amide III and adding a disulphide/aliphatic side-chain motif to carry the rest.
- **Ground the weak motifs with pure references.** Porphyrin, flavin and a true sterol standard are the three highest-value additions to the corpus, and would require a versioned atlas rebuild — which is the only legitimate way to fix the sterol limit.
- **Calibrate the theme confidence** (temperature scaling on the reliability curve) before any downstream consumer treats it as a probability.
- **Add the V6 evaluation to CI.** The tier numbers and per-theme table are cheap to recompute and would catch silent regressions in M or T.
- **Do not fold SERS into any layer below the theme map.** The next phase is an observation model on top of this hierarchy, not a refit of it.

11 · Next phase — Ag-SERS adaptation (not implemented in V6)

V6 is deliberately a Raman hierarchy. Prior work established that on silver the theme ranking collapses to chance (top-1 0.23 against a 0.22 null) and 95 % of analytes are pulled onto a purine attractor that is present in the unspiked blank before any analyte is added. The correct next step is an **observation model** layered on top of the frozen V6 hierarchy — a learned Raman→SERS transfer restricted to the analytes that a Stage-0 detection gate shows are actually measurable on silver — rather than any re-fitting of M, T or the atlas. The V6 contribution to that phase is a clean, non-circular target: a transfer model now has an unambiguous 17-motif / 13-theme representation to map into.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged throughout. All numbers recomputed from results/v6_rebuild/. Reproduce: p01 → p02 → p03 → p05 → p06 → p07.